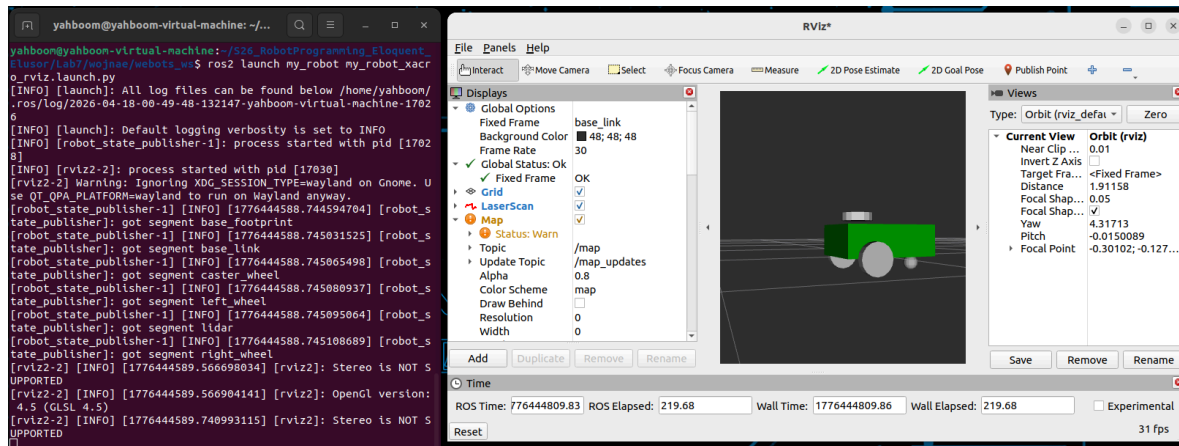


Emma Wojnarowski
662035363

Lab 7 Deliverable

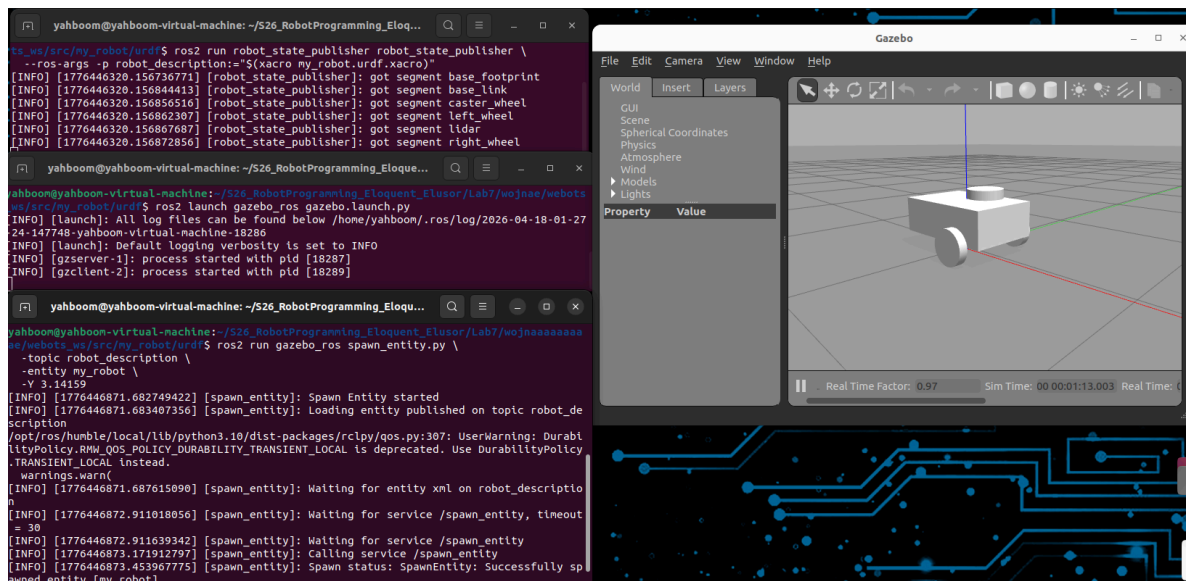
Task 1 Step 10

Here is a screenshot with Rviz and command in the terminal visible.



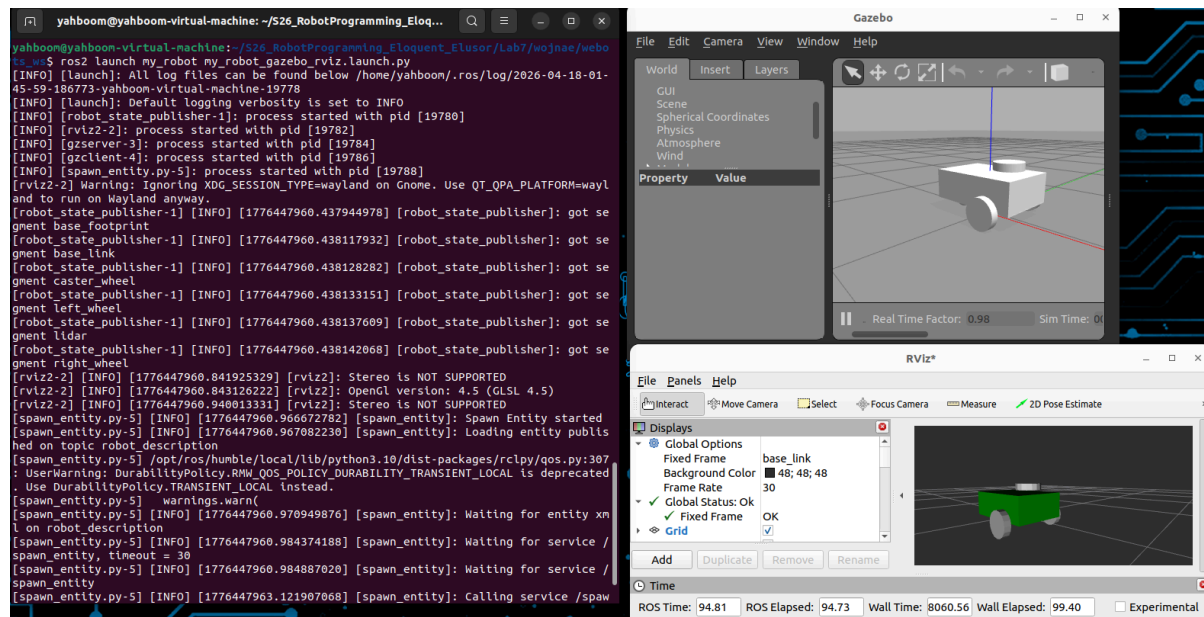
Task 3 Step 2

Here is a screenshot of the terminals and Gazebo window (spawned with yaw because my “front” is technically the “back”). The yaw of 3.14159 flips the base by 180 degrees so I have the correct front side.



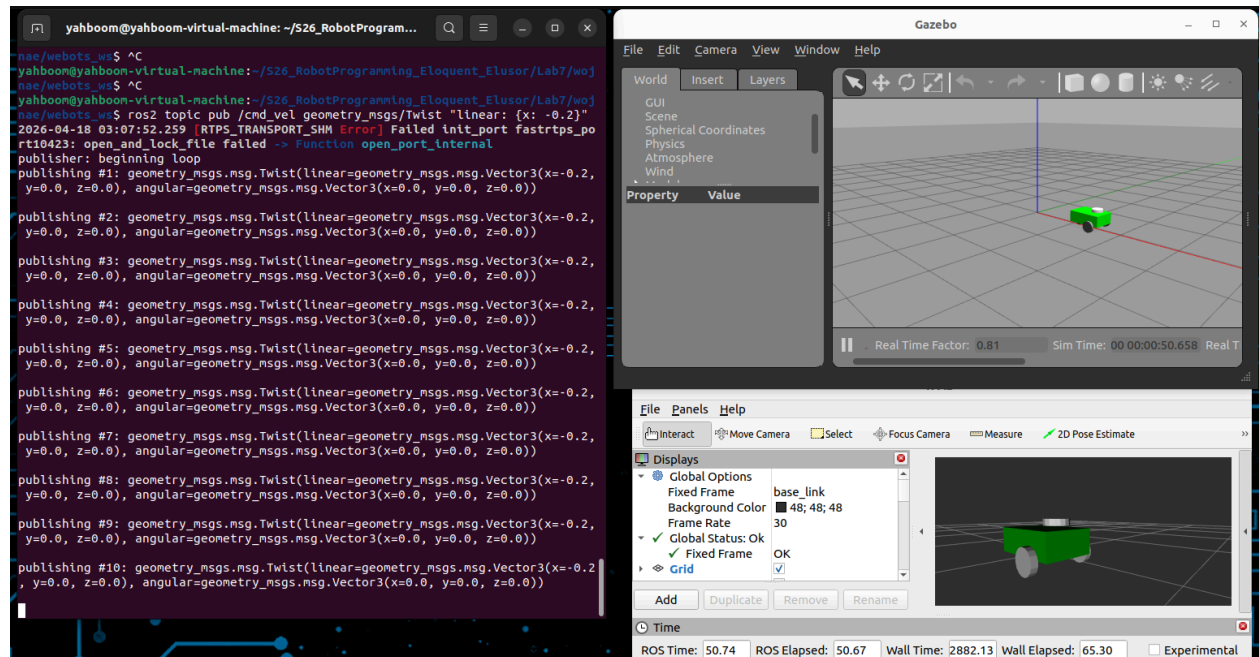
Task 3 Step 7

Here is the completed launch file.



Task 4 Step 8

Here is the Gazebo window after running the launch file (Used “x: -0.2” because of my robot being “flipped” with the front and back. If I had kept it as “x: 0.2” my robot would have been going in the opposite direction).



Completed urdf.xacro File

Here is the completed urdf.xacro file.

```
<?xml version="1.0"?>
<robot name="my_robot" xmlns:xacro="http://www.ros.org/wiki/xacro">

  <!-- ===== XACRO VARIABLES ===== -->
  <xacro:property name="base_length" value="0.6"/>
  <xacro:property name="base_width" value="0.4"/>
  <xacro:property name="base_height" value="0.2"/>
  <xacro:property name="wheel_radius" value="0.1"/>
  <xacro:property name="wheel_length" value="0.05"/>

  <!-- ===== INERTIA MACROS ===== -->
  <xacro:macro name="box_inertia" params="mass lx ly lz xyz rpy">
    <inertial>
      <origin xyz="{xyz}" rpy="{rpy}" />
      <mass value="{mass}" />
      <inertia
        ixx="{(mass / 12.0) * (ly * ly + lz * lz)}"
        ixy="0" ixz="0"
        iyy="{(mass / 12.0) * (lx * lx + lz * lz)}"
        iyz="0"
        izz="{(mass / 12.0) * (lx * lx + ly * ly)}" />
    </inertial>
  </xacro:macro>

  <xacro:macro name="cylinder_inertia" params="mass h r xyz rpy">
    <inertial>
      <origin xyz="{xyz}" rpy="{rpy}" />
      <mass value="{mass}" />
      <inertia
        ixx="{(mass / 12.0) * (3.0 * r * r + h * h)}"
        ixy="0" ixz="0"
        iyy="{(mass / 12.0) * (3.0 * r * r + h * h)}"
        iyz="0"
        izz="{(mass / 2.0) * (r * r)}" />
    </inertial>
  </xacro:macro>

  <xacro:macro name="sphere_inertia" params="mass r xyz rpy">
    <inertial>
      <origin xyz="{xyz}" rpy="{rpy}" />
      <mass value="{mass}" />
      <inertia
        ixx="{(mass * 2.0 / 5.0) * (r * r)}"
        ixy="0" ixz="0"
        iyy="{(mass * 2.0 / 5.0) * (r * r)}"
        iyz="0"
        izz="{(mass * 2.0 / 5.0) * (r * r)}" />
    </inertial>
  </xacro:macro>

  <!-- ===== MATERIALS ===== -->
  <material name="green">
    <color rgba="0 0.6 0 1" />
  </material>

  <material name="white">
    <color rgba="0.8 0.8 0.8 1" />
  </material>

  <material name="gray">
    <color rgba="0.7 0.7 0.7 1" />
  </material>

  <!-- ===== BASE FOOTPRINT ===== -->
  <link name="base_footprint"/>
```

```

<!-- ===== BASE LINK ===== -->
<link name="base_link">
  <visual>
    <geometry>
      <box size="{base_length} {base_width} {base_height}"/>
    </geometry>
    <origin xyz="0 0 {base_height/2.0}" rpy="0 0 0"/>
    <material name="green"/>
  </visual>

  <collision>
    <geometry>
      <box size="{base_length} {base_width} {base_height}"/>
    </geometry>
    <origin xyz="0 0 {base_height/2.0}" rpy="0 0 0"/>
  </collision>

  <xacro:box_inertia
    mass="5.0"
    lx="{base_length}"
    ly="{base_width}"
    lz="{base_height}"
    xyz="0 0 {base_height/2.0}"
    rpy="0 0 0"/>
</link>

<joint name="base_joint" type="fixed">
  <parent link="base_footprint"/>
  <child link="base_link"/>
  <origin xyz="0 0 0.1" rpy="0 0 0"/>
</joint>

<!-- ===== LIDAR ===== -->
<link name="lidar">
  <visual>
    <geometry>
      <cylinder radius="0.1" length="0.05"/>
    </geometry>
    <material name="white"/>
  </visual>

  <collision>
    <geometry>
      <cylinder radius="0.1" length="0.05"/>
    </geometry>
  </collision>

  <xacro:cylinder_inertia
    mass="1.0"
    h="0.05"
    r="0.1"
    xyz="0 0 0"
    rpy="0 0 0"/>
</link>

<joint name="base_lidar_joint" type="fixed">
  <parent link="base_link"/>
  <child link="lidar"/>
  <origin xyz="-0.15 0 0.25" rpy="0 0 0"/>
</joint>

<!-- ===== LEFT WHEEL ===== -->
<link name="left_wheel">
  <visual>
    <geometry>
      <cylinder radius="{wheel_radius}" length="{wheel_length}"/>
    </geometry>
    <material name="gray"/>
  </visual>

```

```

<collision>
  <geometry>
    <cylinder radius="{wheel_radius}" length="{wheel_length}" />
  </geometry>
</collision>

<xacro:cylinder_inertia
  mass="1.0"
  h="{wheel_length}"
  r="{wheel_radius}"
  xyz="0 0 0"
  rpy="0 0 0" />
</link>

<joint name="base_left_wheel_joint" type="continuous">
  <parent link="base_link" />
  <child link="left_wheel" />
  <origin xyz="-0.15 0.25 0" rpy="{-pi / 2.0} 0 0" />
  <axis xyz="0 0 1" />
</joint>

<!-- ===== RIGHT WHEEL ===== -->
<link name="right_wheel">
  <visual>
    <geometry>
      <cylinder radius="{wheel_radius}" length="{wheel_length}" />
    </geometry>
    <material name="gray" />
  </visual>

  <collision>
    <geometry>
      <cylinder radius="{wheel_radius}" length="{wheel_length}" />
    </geometry>
  </collision>

  <xacro:cylinder_inertia
    mass="1.0"
    h="{wheel_length}"
    r="{wheel_radius}"
    xyz="0 0 0"
    rpy="0 0 0" />
</link>

<joint name="base_right_wheel_joint" type="continuous">
  <parent link="base_link" />
  <child link="right_wheel" />
  <origin xyz="-0.15 -0.25 0" rpy="{-pi / 2.0} 0 0" />
  <axis xyz="0 0 1" />
</joint>

<!-- ===== CASTER ===== -->
<link name="caster_wheel">
  <visual>
    <geometry>
      <sphere radius="{wheel_radius/2.0}" />
    </geometry>
    <material name="gray" />
  </visual>

  <collision>
    <geometry>
      <sphere radius="{wheel_radius/2.0}" />
    </geometry>
  </collision>

  <xacro:sphere_inertia
    mass="2.0"
    r="{wheel_radius/2.0}"

```

```

    xyz="0 0 0"
    rpy="0 0 0"/>
</link>

<joint name="base_caster_wheel_joint" type="fixed">
  <parent link="base_link"/>
  <child link="caster_wheel"/>
  <origin xyz="0.25 0 -0.05" rpy="0 0 0"/>
</joint>

<!-- ===== GAZEBO MATERIALS ===== -->
<gazebo reference="base_link">
  <material>Gazebo/Green</material>
</gazebo>

<gazebo reference="left_wheel">
  <material>Gazebo/Grey</material>
</gazebo>

<gazebo reference="right_wheel">
  <material>Gazebo/Grey</material>
</gazebo>

<gazebo reference="caster_wheel">
  <material>Gazebo/Grey</material>
</gazebo>

<gazebo reference="lidar">
  <material>Gazebo/White</material>
</gazebo>

<!-- ===== DIFF DRIVE PLUGIN ===== -->

<gazebo>
  <plugin name="diff_drive_controller"
    filename="libgazebo_ros_diff_drive.so">

    <!-- Update rate in Hz -->
    <update_rate>50</update_rate>

    <!-- Wheels -->
    <left_joint>base_left_wheel_joint</left_joint>
    <right_joint>base_right_wheel_joint</right_joint>

    <!-- Kinematics -->
    <wheel_separation>0.4</wheel_separation>
    <wheel_diameter>0.2</wheel_diameter>

    <!-- Output -->
    <publish_odom>true</publish_odom>
    <publish_odom_tf>true</publish_odom_tf>
    <publish_wheel_tf>true</publish_wheel_tf>

    <odometry_topic>odom</odometry_topic>
    <odometry_frame>odom</odometry_frame>
    <robot_base_frame>base_footprint</robot_base_frame>

    <!-- Input -->
    <command_topic>cmd_vel</command_topic>

  </plugin>
</gazebo>

</robot>

```